Soft Flexible Robotic Manipulators

Under the guidance of
Prof. Anirban Guha

by
Niraj Niraula
24M1723

MASTER OF TECHNOLOGY
DEPARTMENT OF MECHANICAL ENGINEERING

INDIAN INSTITUTE OF TECHNOLOGY, BOMBAY

Content

- Introduction
- Motivation
- History
- Reason of choice over rigid robots
- Material and fabrication
- Actuation method
- Modeling and control
- Sensing technology
- Limitation
- Future direction
- Conclusion

Introduction

- Soft robotics

  Flexible , Deformable material

- Manipulators and gripers

  Full arm , end effector

- stationery Flexible soft manipulator

  Higher degree of freedom

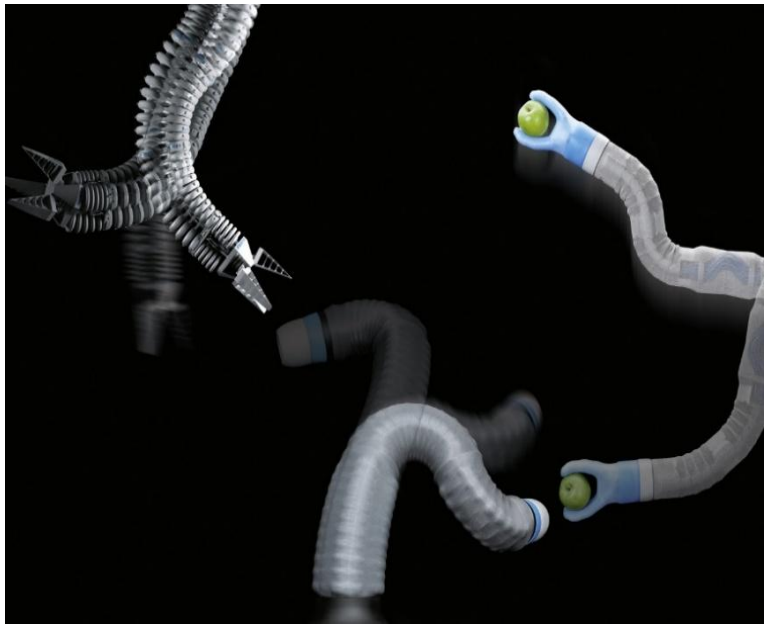


Fig .soft flexible manipulator prototype

[Online]. Available: https://www.tudelft.nl/en/ide/research/technology/soft-robotics.

Motivation

- Traditional rigid robots are:
  - limited in uncertain environments
  - Unsafe for close human interaction
  - Inadequate for handling objects
- Need for robots that are:
  - safe, flexible, adaptive



Fig .Tesla snake charger prototype

https://www.youtube.com/watch?v=uMM0lRfX6YI

Literature review

Historic revolution

- Early 2000s Bio-Inspiration Emerges from octopus, worms
- Early prototypes using elastomers and air chambers
- 2011: Field Defined
- grown immensely over the past decade

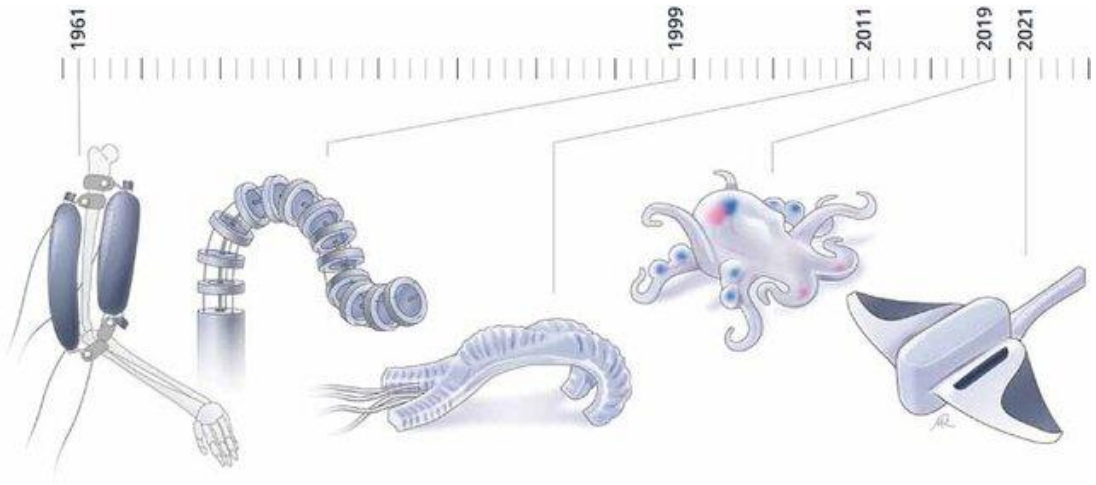


Fig .Revolution of soft robots

Tauber, Falk & Desmulliez, M.P.Y. & Piccin, Olivier & Stokes, Adam. (2023). Perspective for soft robotics: the field's past and future. Bioinspiration & Biomimetics. 18. 10.1088/1748-3190/acbb48.

Choice of Soft Robots over Rigid Robots

- **Safe Human-Robot Interaction**

  safe physical interaction with humans, reduce risk of injury in accidental contact

- **Adoptability and Flexibility**

  navigate in tight, cluttered environments

- **Bio inspired Locomotion and Dexterity**

  natural movements like crawling, swimming, or slithering

Choice of Soft Robots over Rigid Robots

- Delicate Object Handling

  handling without damage

- Durability in Uncertain Environments

  non corrosive, can be used in rescue mission

- Lower Cost and Simpler Design

  Motor ,gear ,metals removed suitable for small business and labs

Choice of Soft Robots over Rigid Robots

- Built-In Environmental Compliance

natural shape adaptation with out complex sensor



Fig. shape adoption according to shape

Hussain, Irfan Al-Ketan, Oraib Design and prototyping soft–rigid tendon-driven modular grippers using interpenetrating phase composites materials,2020

Materials and Fabrication Techniques

- Silicone Elastomers $-[Si-O]-_n$

  most commonly used, mechanical flexibility, chemical stability, and biocompatibility

- Hydrogels and Smart Materials

  jelly like material ⟶ swell, contract and SMP,EAP

- 3D Printing and Additive Manufacturing

  revolutionized the fabrication, fused deposition modeling (FDM), stereolithography (SLA)

Yap, Y. L., Sing, S. L., & Yeong, W. Y. (2020). A review of 3D printing processes and materials for soft robotics. Rapid Prototyping Journal, 26(8), 1345-1361.

Materials and Fabrication Techniques

Molding and Casting

Molding ⟶ internal channels and chambers

casting ⟶ Anatomical shape

Actuation method

- Fluid driven

gas , liquid

gas driven ⟶ pollution free , widely used

pressure regulation principle

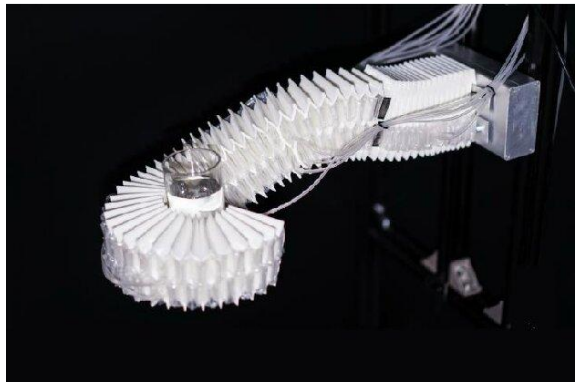


Fig. pneumatic soft robot

Hao, Jiang & Liu, Xinghua & Chen, Xiaotong & Wang, Zhanchi & Jin, Yusong & Chen, Xiaoping. (2016). Design and simulation analysis of a soft manipulator based on honeycomb pneumatic networks. 350-356. 10.1109/ROBIO.2016.7866347.

Actuation method

Cable driven

actuation by cable tension

motor ⟶ pulley ⟶ cable tension

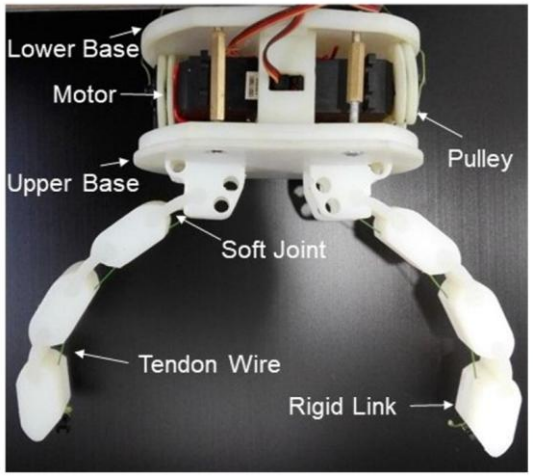


Fig. cable driven soft robot

EAP driven

actuation by electrostatic attraction

electronic type, ionic type

Hussain, Irfan Al-Ketan, Oraib Design and prototyping soft–rigid tendon-driven modular grippers using interpenetrating phase composites materials,2020

Actuation method

- Shape memory materials-driven

actuation -shape restore at a specific temperature

Shape memory alloy and shape memory polymer

↓ nitinol

↓ polyurethane



Vid. Shape memory alloy

https://www.youtube.com/shorts/HJUI5JYeLCM

Actuation method

Electromagnetic-driven mode

actuation by interaction ferromagnetic and magnetic field

Hybrid drive Mode

combination of two or more different drive modes

improve load capacity ⟶ increase control accuracy

example. Cable driven weak in pressure good in tenson ⟶ cable+liquid

Chen, Xiaoqian & Zhang, Xiang & Huang, Yiyong & Cao, Lu & Liu, Jinguo. (2021). A review of soft manipulator research, applications, and opportunities

Modeling and Control

Kinematic Modeling

Piecewise Constant Curvature (PCC)

divided into segments

each section bends in constant curvature

length of segment ,angle of curvature,

plane inclination

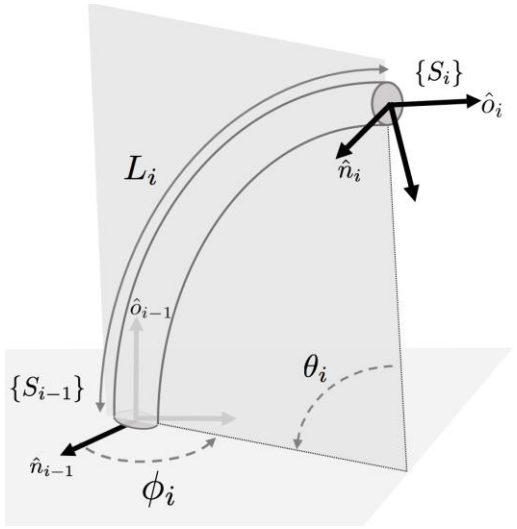


Fig. kinematic representation Of ith segment

Till, John & Aloi, Vincent & Rucker, D.. (2019). Real-Time Dynamics of Soft and Continuum Robots based on Cosserat-Rod Models. The International Journal of Robotics Research. 38. 723-746. 10.1177/0278364919842269.

Modeling and Control

Kinematic Modeling

Cosserat Rod Theory

Robotic arm modeled as continuous rod

driven by curve ⟶ differential equation

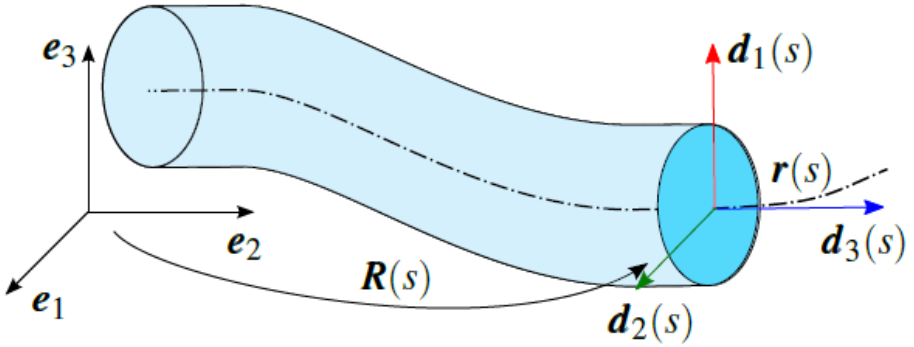


Fig. CRT model

Dynamic modeling

Euler–Bernoulli beam theory ⟶ small load

Cosserat theory ⟶ big load

mass-spring-damper systems ⟶ simplified approximations

Katzschmann, Robert & Della Santina, Cosimo & Toshimitsu, Yasunori & Bicchi, Antonio & Rus, Daniela. (2019). Dynamic Motion Control of Multi-Segment Soft Robots Using Piecewise Constant Curvature Matched with an Augmented Rigid Body Model. 10.1109/ROBOSOFT.2019.8722799.

Modeling and Control

Finite Element Modeling (FEM)

complex geometries , material nonlinearities

ANSYS, **Abaqus**, **SOFA**

may not be suitable for real time control

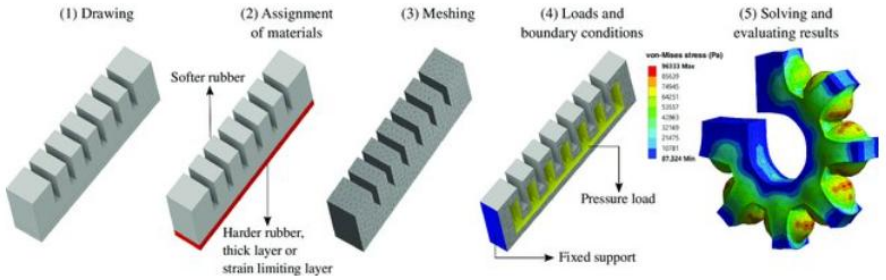


Fig. FEM of soft robot

Data-Driven and Machine Learning Models

popular in soft robot

Experimental data ⟶ train model ⟶ predict

Xavier, Matheus & Fleming, Andrew & Yong, Yuen Kuan. (2020). Finite Element Modeling of Soft Fluidic Actuators: Overview and Recent Developments

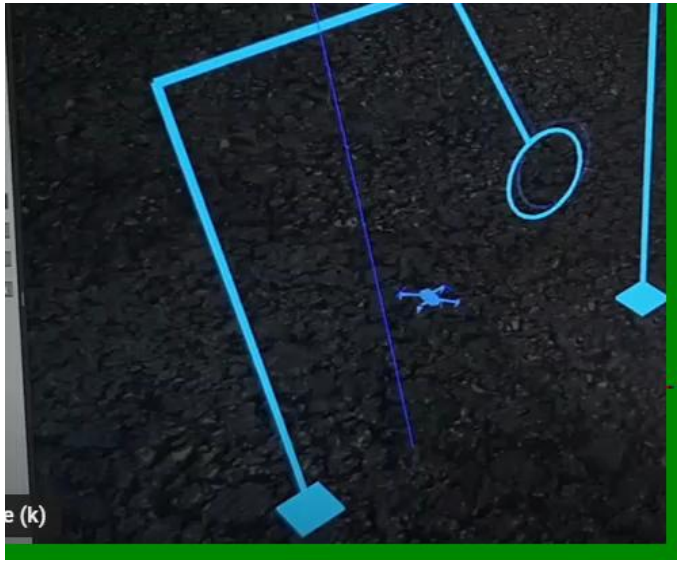
Modeling and Control

Control strategy

open loop ⟶ simple, limited accuracy, used for demonstration

closed loop ⟶ Uses sensors (e.g., pressure, strain, vision) to adjust inputs in real time

learning base model predictive control ⟶



Bruder, D., Gillespie, B., Remy, C. D., & Vasudevan, R. (2019). Modeling and control of soft robots using the koopman operator and model predictive control. *arXiv preprint arXiv:1902.02827*
https://www.youtube.com/watch?v=OSPpioiGKPo.

Sensing Technology

Tactile sensing

ability to detect and respond on physical interaction

Ex. grip without slippage and damage

resistive , capacitive ,piezoelectric

Proprioception

Ability to sense own state(position , velocity)

stretchable strain sensors, distributed encoders

feedback on bending, stretching

each segment of arm

Sensing Technology

- **Vision system**

  perceives environment ⟶ adjust action based on visuals

  Object recognition, Localization

- **Force and Pressure Sensing**

  detect the forces applied during interaction (Pneumatic , capacitive ,piezoelectric)

- **Environmental Sensing**

perceive external conditions, temperature, humidity, obstacles

Hegde C, Su J, Tan JMR, He K, Chen X, Magdassi S. Sensing in Soft Robotics. ACS Nano. 2023 Aug

Challenges and Limitations in Soft Robotic Manipulators

Material Limitations

Material fatigue, limited durability, limited strength to weight ratio

Actuation Challenges

limited control accuracy, complex actuation mechanism, limited force output

Control Complexity

Non linear behavior ⟶ limited control algorithm

large number of variable ⟶ difficult to design control

Challenges and Limitations in Soft Robotic Manipulators

Sensing Challenges

effective sensing is crucial

limited flexible sensors,

Feedback accuracy

Durability and Longevity

Deformation-induced wear, Environmental factors, Maintenance challenges

Challenges and Limitations in Soft Robotic Manipulators

Cost and Scalability

High material costs, Complex fabrication processes

Scaling production

Wehner, M., Truby, R., Fitzgerald, D. *et al.* An integrated design and fabrication strategy for entirely soft, autonomous robots

Future Directions in Soft Robotic Manipulators

- Advanced Materials Development

  self healing material , electro active and shape memory polymers

- Actuation Innovation

  High-performance soft actuators,Maniaturization

- Integration with Artificial Intelligence (AI) and Machine Learning (ML)

  Autonomous task planning

Future Directions in Soft Robotic Manipulators

- **Bio inspired Designs**

  Soft actuators inspired by biological muscles, locomotion based on animal movement

- **Hybrid Soft-Rigid Robots**

  Enhanced load-bearing capacity, Improved dexterity and control

- **Industry collaboration**

  cobots , medical robot, Agriculture and exploration

Conclusion

- Rapid revolution
- Mainly (pneumatic ,cable driven , EAP) actuation are used
- Implementing AI
- 3D printed soft robots
- Bio mimicking

Future work

To used this knowledge to research of Robotic Solutions for Electric Vehicle Charge-Rejuvenating Centres



Thank you !